

7.4 Optimization Using Quadratic Forms

Quadratic forms arise in various problems in which the maximum or minimum value of some quantity is required. In this section we will discuss some problems of this type.

Constrained Extremum Problems

The primary goal of this section is to find the maximum and minimum values of a quadratic form $\mathbf{x}^T A \mathbf{x}$ subject to the constraint $\|\mathbf{x}\| = 1$. Such problems arise in a wide variety of applications.

To visualize this geometrically in R^2 : view $z = \mathbf{x}^T A \mathbf{x}$ as a surface in xyz -coordinates, and $\|\mathbf{x}\| = 1$ as the unit circle in the xy -plane. Finding the max and min of z on $\|\mathbf{x}\| = 1$ amounts to finding the highest and lowest points on the intersection of the surface with the right circular cylinder over the unit circle (Figure 7.4.1).

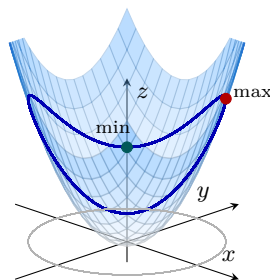


Figure 1: The constrained max and min of $\mathbf{x}^T A \mathbf{x}$ on $\|\mathbf{x}\| = 1$ are the highest and lowest points on the curve where the surface meets the unit cylinder.

Theorem 7.4.1 — Constrained Extremum Theorem

Let A be a symmetric $n \times n$ matrix whose eigenvalues in decreasing order are $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. Then:

- (a) $\mathbf{x}^T A \mathbf{x}$ has a maximum value of λ_1 and a minimum value of λ_n , both attained on the set $\|\mathbf{x}\| = 1$.
- (b) The maximum value of $\mathbf{x}^T A \mathbf{x}$ occurs at a unit eigenvector corresponding to λ_1 .
- (c) The minimum value of $\mathbf{x}^T A \mathbf{x}$ occurs at a unit eigenvector corresponding to λ_n .

Proof. By the Principal Axes Theorem (7.3.1), there is an orthogonal change of variable $\mathbf{x} = P\mathbf{y}$ with the column vectors of P ordered so that $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$, giving

$$\mathbf{x}^T A \mathbf{x} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2 \quad (6)$$

Since P is orthogonal it is length-preserving, so $\|\mathbf{y}\| = \|\mathbf{x}\| = 1$, i.e.,

$$y_1^2 + y_2^2 + \cdots + y_n^2 = 1 \quad (7)$$

From (7) and $\lambda_1 \geq \cdots \geq \lambda_n$:

$$\lambda_n = \lambda_n(y_1^2 + \cdots + y_n^2) \leq \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 \leq \lambda_1(y_1^2 + \cdots + y_n^2) = \lambda_1$$

and by (6), $\lambda_n \leq \mathbf{x}^T A \mathbf{x} \leq \lambda_1$ for all $\|\mathbf{x}\| = 1$.

If $\mathbf{x}$ is a unit eigenvector corresponding to λ_1 , then $\mathbf{x}^T A \mathbf{x} = \mathbf{x}^T (\lambda_1 \mathbf{x}) = \lambda_1 \|\mathbf{x}\|^2 = \lambda_1$, so the maximum λ_1 is achieved. Similarly, if $\mathbf{x}$ is a unit eigenvector corresponding to λ_n , then $\mathbf{x}^T A \mathbf{x} = \lambda_n$, so the minimum λ_n is achieved. ■

Remark. The condition $\|\mathbf{x}\| = 1$ is called a **constraint**, and the maximum or minimum value is called a **constrained extremum**. The constraint can also be expressed as $\mathbf{x}^T \mathbf{x} = 1$ or as $x_1^2 + x_2^2 + \cdots + x_n^2 = 1$.

EXAMPLE 1 | Finding Constrained Extrema

Find the maximum and minimum values of $z = 5x^2 + 5y^2 + 4xy$ subject to $x^2 + y^2 = 1$, and the values of x and y at which they occur.

SOLUTION: In matrix notation,

$$z = \mathbf{x}^T A \mathbf{x} = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 5 & 2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

The eigenvalues of A are $\lambda_1 = 7$ and $\lambda_2 = 3$ (verify), with corresponding unit eigenvectors

$$\lambda_1 = 7 : \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, \quad \lambda_2 = 3 : \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \quad (1)$$

Thus the constrained extrema are

$$\begin{aligned} \text{constrained maximum: } z &= 7 \text{ at } (x, y) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \\ \text{constrained minimum: } z &= 3 \text{ at } (x, y) = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \end{aligned}$$

Remark. The negatives of the eigenvectors in (1) are also unit eigenvectors, so the constrained maximum $z = 7$ also occurs at $(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$ and the constrained minimum $z = 3$ at $(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$.

EXAMPLE 2 | A Constrained Extremum Problem

A rectangle is to be inscribed in the ellipse $4x^2 + 9y^2 = 36$ with sides parallel to the coordinate axes. Find the nonnegative values of x and y producing the rectangle with maximum area.

SOLUTION: The area of the inscribed rectangle is $z = 4xy$. Rewrite the constraint as

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$$

and introduce new variables x_1, y_1 via $x = 3x_1$, $y = 2y_1$, so the constraint becomes $x_1^2 + y_1^2 = 1$ and the problem is to maximize

$$z = 4(3x_1)(2y_1) = 24x_1y_1$$

Writing $z = 24x_1y_1 = \mathbf{x}_1^T A \mathbf{x}_1$ with

$$A = \begin{bmatrix} 0 & 12 \\ 12 & 0 \end{bmatrix}$$

the largest eigenvalue of A is $\lambda = 12$ (verify), with unit eigenvector $[1/\sqrt{2}, 1/\sqrt{2}]^T$ (nonneg entries). The maximum area is $z = 12$, occurring at

$$x = 3x_1 = \frac{3}{\sqrt{2}}, \quad y = 2y_1 = \frac{2}{\sqrt{2}}$$

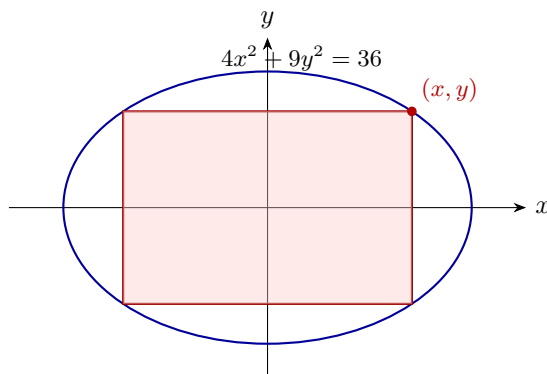


Figure 2: A rectangle with maximum area inscribed in the ellipse $4x^2 + 9y^2 = 36$.

Constrained Extrema and Level Curves

The **level curves** of $f(x, y)$ are the curves $f(x, y) = k$. The level curves of a quadratic form $\mathbf{x}^T A \mathbf{x}$ on $\mathbb{R}^2$ have equations $\mathbf{x}^T A \mathbf{x} = k$. The maximum and minimum values of $\mathbf{x}^T A \mathbf{x}$ on $\|\mathbf{x}\| = 1$ are the largest and smallest values of k for which the level curve $\mathbf{x}^T A \mathbf{x} = k$ intersects the unit circle. Typically, the extreme values correspond to level curves that just *touch* (are tangent to) the unit circle.

EXAMPLE 3 | Example 1 Revisited Using Level Curves

In Example 1 the constrained maximum is $z = 7$ at $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ and $(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$, and the constrained minimum is $z = 3$ at $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ and $(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$. Geometrically, the level curve $5x^2 + 5y^2 + 4xy = 7$ just touches the unit circle at the two maximum points (3), and the level curve $5x^2 + 5y^2 + 4xy = 3$ just touches the unit circle at the two minimum points (4) — see Figure 7.4.5.

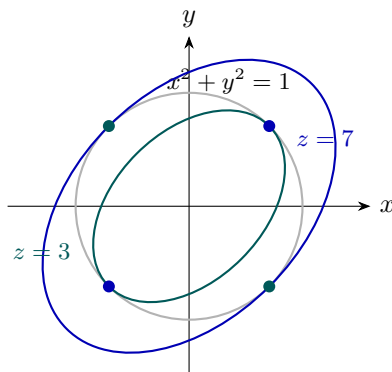


Figure 3: Level curves of $z = 5x^2 + 5y^2 + 4xy$ tangent to the unit circle at the constrained extrema.

Relative Extrema of Functions of Two Variables

(Calculus Required)

We now show how quadratic forms can be used to study characteristics of real-valued functions of two variables.

Recall that if $f(x, y)$ has first-order partial derivatives, its relative maxima and minima occur at **critical points** where

$$f_x(x, y) = 0 \quad \text{and} \quad f_y(x, y) = 0$$

The behavior of f at a critical point (x_0, y_0) is determined by the sign of

$$D(x, y) = f(x, y) - f(x_0, y_0) \tag{5}$$

at nearby points $(x, y) \neq (x_0, y_0)$:

- $D > 0$ near (x_0, y_0) : f has a **relative minimum** at (x_0, y_0)
- $D < 0$ near (x_0, y_0) : f has a **relative maximum** at (x_0, y_0)
- D takes both signs: f has a **saddle point** at (x_0, y_0)

The following theorem, proved in calculus, makes it possible to analyze critical points using second-order partial derivatives.

Theorem 7.4.2 — Second Derivative Test

Suppose (x_0, y_0) is a critical point of $f(x, y)$ and f has continuous second-order partial derivatives in some circular region centered at (x_0, y_0) . Then:

- (a) f has a relative minimum at (x_0, y_0) if

$$f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0) > 0 \quad \text{and} \quad f_{xx}(x_0, y_0) > 0$$

- (b) f has a relative maximum at (x_0, y_0) if

$$f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0) > 0 \quad \text{and} \quad f_{xx}(x_0, y_0) < 0$$

- (c) f has a saddle point at (x_0, y_0) if

$$f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0) < 0$$

- (d) The test is inconclusive if $f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0) = 0$.

The expression $f_{xx}f_{yy} - f_{xy}^2$ can be expressed using the **Hessian matrix** (named for the German mathematician Ludwig Otto Hesse, 1811–1874):

$$H(x, y) = \begin{bmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{xy}(x, y) & f_{yy}(x, y) \end{bmatrix}$$

Then

$$\det[H(x_0, y_0)] = \begin{vmatrix} f_{xx}(x_0, y_0) & f_{xy}(x_0, y_0) \\ f_{xy}(x_0, y_0) & f_{yy}(x_0, y_0) \end{vmatrix} = f_{xx}f_{yy} - f_{xy}^2|_{(x_0, y_0)}$$

Theorem 7.4.2 can now be reformulated in terms of the definiteness of $H(x_0, y_0)$.

Theorem 7.4.3 — Hessian Form of the Second Derivative Test

Suppose (x_0, y_0) is a critical point of $f(x, y)$ and f has continuous second-order partial derivatives in some circular region centered at (x_0, y_0) . If $H(x_0, y_0)$ is the Hessian of f at (x_0, y_0) , then:

- (a) f has a relative minimum at (x_0, y_0) if $H(x_0, y_0)$ is positive definite.
- (b) f has a relative maximum at (x_0, y_0) if $H(x_0, y_0)$ is negative definite.
- (c) f has a saddle point at (x_0, y_0) if $H(x_0, y_0)$ is indefinite.
- (d) The test is inconclusive otherwise.

Proof of (a). If $H(x_0, y_0)$ is positive definite, then Theorem 7.3.4 implies all principal submatrix determinants are positive. In particular,

$$\det[H(x_0, y_0)] = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0) > 0$$

and $\det[f_{xx}(x_0, y_0)] = f_{xx}(x_0, y_0) > 0$, so f has a relative minimum at (x_0, y_0) by Theorem 7.4.2(a). ■

(The proofs of parts (b)–(d) are analogous and are left as exercises.)

EXAMPLE 4 | Using the Hessian to Classify Relative Extrema

Find the critical points of $f(x, y) = \frac{1}{3}x^3 + xy^2 - 8xy + 3$ and use the eigenvalues of the Hessian matrix at those points to classify each as a relative maximum, relative minimum, or saddle point.

SOLUTION: The first and second partial derivatives are

$$f_x = x^2 + y^2 - 8y, \quad f_y = 2xy - 8x = 2x(y - 4), \quad f_{xx} = 2x, \quad f_{yy} = 2x, \quad f_{xy} = 2y - 8$$

Setting $f_y = 0$: either $x = 0$ or $y = 4$.

- $x = 0$: $f_x = y^2 - 8y = y(y - 8) = 0$ gives $y = 0$ or $y = 8$.
- $y = 4$: $f_x = x^2 + 16 - 32 = x^2 - 16 = 0$ gives $x = \pm 4$.

The four critical points are $(0, 0)$, $(0, 8)$, $(4, 4)$, $(-4, 4)$. The Hessian matrix is

$$H(x, y) = \begin{bmatrix} 2x & 2y - 8 \\ 2y - 8 & 2x \end{bmatrix}$$

Evaluating at the critical points:

$$H(0, 0) = \begin{bmatrix} 0 & -8 \\ -8 & 0 \end{bmatrix}, \quad H(0, 8) = \begin{bmatrix} 0 & 8 \\ 8 & 0 \end{bmatrix}, \quad H(4, 4) = \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix}, \quad H(-4, 4) = \begin{bmatrix} -8 & 0 \\ 0 & -8 \end{bmatrix}$$

Critical Point (x_0, y_0)	λ_1	λ_2	Classification
$(0, 0)$	8	-8	Saddle point
$(0, 8)$	8	-8	Saddle point
$(4, 4)$	8	8	Relative minimum
$(-4, 4)$	-8	-8	Relative maximum

Exercise Set 7.4

In Exercises 1–4, find the maximum and minimum values of the quadratic form subject to the constraint $x^2 + y^2 = 1$, and determine the values of x and y at which the maximum and minimum occur.

1. $5x^2 - y^2$

2. xy

3. $3x^2 + 7y^2$

In Exercises 5–6, find the maximum and minimum values of the quadratic form subject to the constraint $x^2 + y^2 + z^2 = 1$, and determine the values of x , y , z at which the maximum and minimum occur.

5. $9x^2 + 4y^2 + 3z^2$

In Exercises 7–8, use the method of Example 2 to find the maximum and minimum values of the given quadratic form subject to the stated constraint.

7. xy ; $4x^2 + 8y^2 = 16$

8. $x^2 + xy + 2y^2$; $x^2 + 3y^2 = 16$

In Exercises 9–10, draw the unit circle and the level curves of the quadratic form corresponding to the constrained maximum and minimum. Show that the unit circle intersects each of these curves in exactly two places, label those intersection points, and verify that the constrained extrema occur at those points.

10. xy

In Exercises 11–12, (a) show that the given function has critical points at the stated locations, and (b) use the Hessian form of the second derivative test to show the stated classification.

11. $f(x, y) = 4xy - x^4 - y^4$

a. Critical points at $(0, 0)$, $(1, 1)$, and $(-1, -1)$.

b. f has relative maxima at $(1, 1)$ and $(-1, -1)$, and a saddle point at $(0, 0)$.

12. $f(x, y) = x^3 - 6xy - y^3$

a. Critical points at $(0, 0)$ and $(-2, 2)$.

b. f has a relative maximum at $(-2, 2)$ and a saddle point at $(0, 0)$.

In Exercises 13–16, find the critical points of f , if any, and classify them as relative maxima, relative minima, or saddle points.

13. $f(x, y) = x^3 - 3xy - y^3$
15. $f(x, y) = x^2 + 2y^2 - x^2y$
16. $f(x, y) = x^3 + y^3 - 3x - 3y$
17. A rectangle whose center is at the origin and whose sides are parallel to the coordinate axes is to be inscribed in the ellipse $x^2 + 25y^2 = 25$. Use the method of Example 2 to find nonnegative values of x and y that produce the inscribed rectangle with maximum area.
18. Suppose $\mathbf{x}$ is a unit eigenvector of a matrix A corresponding to an eigenvalue $\lambda = 2$. What is the value of $\mathbf{x}^T A \mathbf{x}$?

Working with Proofs

21. Suppose A is an $n \times n$ symmetric matrix and $q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$, where $\mathbf{x}$ is a vector in R^n expressed in column form. What can you say about the value of q if $\mathbf{x}$ is a unit eigenvector corresponding to an eigenvalue λ of A ?
22. Prove: If $\mathbf{x}^T A \mathbf{x}$ is a quadratic form whose minimum and maximum values subject to the constraint $\|\mathbf{x}\| = 1$ are m and M , respectively, then for each number c in the interval $m \leq c \leq M$ there is a unit vector $\mathbf{x}_c$ such that $\mathbf{x}_c^T A \mathbf{x}_c = c$.

[Hint: In the case $m < M$, let $\mathbf{u}_m$ and $\mathbf{u}_M$ be unit eigenvectors of A such that $\mathbf{u}_m^T A \mathbf{u}_m = m$ and $\mathbf{u}_M^T A \mathbf{u}_M = M$, and let

$$\mathbf{x}_c = \sqrt{\frac{M-c}{M-m}} \mathbf{u}_m + \sqrt{\frac{c-m}{M-m}} \mathbf{u}_M$$

Show that $\mathbf{x}_c^T A \mathbf{x}_c = c$.]